

2025/1/28

Wolfgang Walter, p. 165

$$\dot{x} = A(t)x \quad \begin{matrix} \nearrow n \times n \end{matrix}$$

Prop (a) If $x(t)$ is a solution, and $x(t_0) = 0$ for some $t_0 \in \mathbb{R}$, then $x(t) \equiv 0$ (i.e., $x(t) = 0 \forall t$)

(b) A set of k solutions $x_1(t), \dots, x_k(t)$ is linearly independent if

$$c_1 x_1(t) + \dots + c_k x_k(t) = 0 \Rightarrow c_1 = c_2 = \dots = c_k = 0$$

$\neq$
zero sol'n

o/w $x_1(t), \dots, x_k(t)$ is called linearly dependent.

If the eqn

$$c_1 x_1(t) + \dots + c_k x_k(t) = 0$$

holds at some point t_0 , then it holds identically for all t . (by (a) & linearity)

(c) ignore.

(d) for $k > n$, any k sol'n's $x_1(t), \dots, x_k(t)$ are linearly dependent

(e) $\exists$ n linearly independent solutions $x_1(t), \dots, x_n(t)$ to the eqn $\dot{x} = A(t)x$.

Any such set of n linearly independent sol'n's is called a fundamental set/system of sol'n's

If $x_1(t), \dots, x_n(t)$ is a fundamental set of sol'n's, then any sol'n $x(t) \in \dot{x} = A(t)x$ can be uniquely written as a linear combination

$$x(t) = c_1 x_1(t) + \dots + c_n x_n(t)$$

(f) A system of sol'n $x_1(t), \dots, x_n(t)$ can be assembled into an $n \times n$ sol'n matrix

$$\underline{\Psi}(t) = \begin{bmatrix} | & & | \\ x_1(t) & \dots & x_n(t) \\ | & & | \end{bmatrix}$$

then we have a matrix ODE for $\underline{\Psi}$

$$\dot{\underline{\Psi}} = A(t) \underline{\Psi}$$

According to (b),

$\underline{\Psi}(t)$ is singular for a particular $t_0 \in I \subseteq \mathbb{R}$ ^{interval}

$\Leftrightarrow \underline{\Psi}(t)$ is singular $\forall t \in I$,

Def (Wronskian) If $\underline{X}(t) = \begin{bmatrix} | & & | \\ x_1(t) & \dots & x_n(t) \\ | & & | \end{bmatrix}$ is a solution to $\dot{\underline{\Psi}} = A(t) \underline{\Psi}$, then the its determinant $W(t) := \det \underline{X}(t)$ is called the Wronskian (determinant) of $\underline{X}(t)$.

Thm If $A(t)$ is continuous in $J \subseteq \mathbb{R}$, then

$$\dot{W} = \underbrace{\text{tr}(A(t))}_{\text{scalar ODE}} W \quad \forall t \in J$$

$$w / \text{tr}(A(t)) = \sum_{i=1}^n A_{ii}(t)$$

$$\text{So } W(t) = W(t_0) \exp \left(\underbrace{\int_{t_0}^t \text{tr}(A(s)) ds}_{> 0} \right)$$

Cor The Wronskian $W(t)$ is either $\equiv 0$ or $\neq 0$ in J

$\forall t \in J, W(t) \neq 0 \Leftrightarrow \Psi(t)$ is fundamental matrix

$\exists t \in J, W(t) \neq 0$

(col's of $\Psi(t)$ form a basis)

Derivative of determinant

$$x(t), y(t), z(t) \in \mathbb{R}^3 \quad \forall t$$

$$\frac{d}{dt} \det(x(t), y(t), z(t)) \quad \leftarrow (x(t) \times y(t)) \cdot z(t)$$

$$= \det(\dot{x}(t), y(t), z(t))$$

$$+ \det(x(t), \dot{y}(t), z(t))$$

$$+ \det(x(t), y(t), \dot{z}(t))$$

Let $A: \mathbb{R} \rightarrow M_n(\mathbb{R})$

$$t \mapsto (a_{ij}(t))_{i,j=1}^n \quad \left(A(t) \in M_n(\mathbb{R}) \quad \forall t \in \mathbb{R} \right)$$

& $\det: M_n(\mathbb{R}) \rightarrow \mathbb{R}$

$$A(t) \mapsto \det(A(t))$$

$$\mathbb{R} \xrightarrow{A(\cdot)} M_n(\mathbb{R}) \xrightarrow{\det} \mathbb{R}$$


we want to calculate

$$\frac{d}{dt} \det(A(t)) = \frac{d}{dt} (\det \circ A)(t)$$

later on this.

Chain Rule

$$D(f \circ g)(x) = Df(g(x)) \overset{\text{composition}}{\downarrow} \text{of linear maps} Dg(x)$$

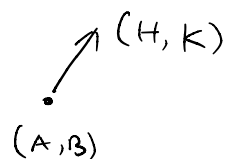
Directional derivative

Consider a bilinear function

$$f: M_{n \times n}(\mathbb{R}) \times M_{n \times p}(\mathbb{R}) \rightarrow M_{n \times p}(\mathbb{R})$$

$$(A, B) \mapsto AB = f(A, B)$$

The directional derivative of f
at (A, B) in the direction (H, K)



$$(A, B) \nearrow (H, K)$$

$$Df(A, B) \cdot (H, K) := \left. \frac{d}{dt} \right|_{t=0} f(A+tH, B+tK)$$

$\uparrow$
 in the direction
 (NOT dot product)

$$= \left. \frac{d}{dt} \right|_{t=0} (A+tH)(B+tK)$$

$$= \left. \frac{d}{dt} (AB + tAK + tHB + t^2HK) \right|_{t=0}$$

$$= AK + HB + 2tHK \Big|_{t=0}$$

$$= AK + HB$$

Observe that for fixed (A, B)

$$Df(A, B): M_{n \times n}(\mathbb{R}) \times M_{n \times p}(\mathbb{R}) \rightarrow M_{n \times p}(\mathbb{R})$$

$$(H, K) \mapsto AK + HB$$

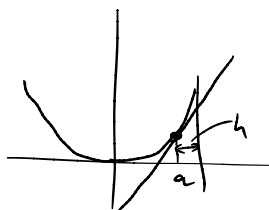
is a linear map.

$$f: \mathbb{R} \rightarrow \mathbb{R}, \quad x \mapsto x^2$$

$$Df(a) = f'(a) = 2a: \mathbb{R} \rightarrow \mathbb{R}$$

$$h \mapsto 2ah$$

is linear



$$Df(a) \cdot h = 2a \cdot h = 2ah$$

ex Consider $g: M_n(\mathbb{R}) \longrightarrow M_n(\mathbb{R})$

$$A \longmapsto A^2$$

By def'n

$$Dg(A) \cdot H = \left. \frac{d}{dt} \right|_{t=0} (A + tH)^2$$

$$= \left. \frac{d}{dt} \right|_{t=0} (A + tH)(A + tH)$$

$$= \left. \frac{d}{dt} \right|_{t=0} (A^2 + tHA + tAH + t^2H^2)$$

$$= HA + AH$$

Observe that that map (for fixed A)

$$Dg(A): M_n(\mathbb{R}) \longrightarrow M_n(\mathbb{R})$$

$$H \longmapsto HA + AH$$

is linear.

In the case $n=1$, $a \in M_1(\mathbb{R}) = \mathbb{R}$, $g(a) = a^2$

$$Dg(a) \cdot h = ah + ha = 2ah$$

$$\text{So } Dg(a): \mathbb{R} \longrightarrow \mathbb{R}$$

$$h \longmapsto 2ah$$

$\uparrow$

$$g'(a) = 2a$$

Let $A \in M_n(\mathbb{R})$ be invertible, i.e., $A \in GL_n(\mathbb{R})$

Consider the inversion map

$$\begin{aligned} i : GL_n(\mathbb{R}) \subseteq M_n(\mathbb{R}) &\longrightarrow GL_n(\mathbb{R}) \subseteq M_n(\mathbb{R}) \\ A &\longmapsto A^{-1} \end{aligned}$$

By def'n,

$$\begin{aligned} Di(A) \cdot H &= \left. \frac{d}{dt} \right|_{t=0} i(A + tH) \\ &= \left. \frac{d}{dt} \right|_{t=0} (A + tH)^{-1} \\ &= \left. \frac{d}{dt} \right|_{t=0} (A(I_n + tA^{-1}H))^{-1} \\ &= \left(\left. \frac{d}{dt} \right|_{t=0} (I_n + tA^{-1}H) \right) A^{-1} \\ &= \left(\left. \frac{d}{dt} \right|_{t=0} (I_n + \underline{\underline{tA^{-1}H}} + t^2(\dots) + \dots) \right) \underline{\underline{A^{-1}}} \\ &= -A^{-1}H A^{-1} \end{aligned}$$

Observe that the map

$$\begin{aligned} Di(A) : M_n(\mathbb{R}) &\longrightarrow M_n(\mathbb{R}) \\ H &\longmapsto -A^{-1}H A^{-1} \end{aligned}$$

is linear (in H)

In the case $n=1$, $i(a) = a^{-1} = \frac{1}{a}$, $a \neq 0$

$$\text{So } Di(a) \cdot h = -a^{-1} h a^{-1} = -\frac{1}{a^2} h$$

$\left(\frac{1}{a} \right)'$

Consider an n -linear map

$$f: E_1 \times E_2 \times \dots \times E_n \longrightarrow F$$

$$(x_1, \dots, x_n) \mapsto f(x_1, \dots, x_n)$$

where $E_1, \dots, E_n, F$ are vector spaces.

For any fixed k , $1 \leq k \leq n$,
and fixed $x_1, \dots, x_{k-1}, x_{k+1}, \dots, x_n$,

$$f(x_1, \dots, x_{k-1}, \cdot, x_{k+1}, \dots, x_n): E_k \rightarrow F$$

is linear

ex $\det: \underbrace{\mathbb{R}^n \times \dots \times \mathbb{R}^n}_{\cong M_n(\mathbb{R})} \rightarrow \mathbb{R}$ is n -linear.

$$\text{Given } f: E_1 \times E_2 \times \dots \times E_n \longrightarrow F$$

$$(x_1, \dots, x_n) \mapsto f(x_1, \dots, x_n)$$

n -linear, then

$$Df(x_1, \dots, x_n) \cdot (h_1, \dots, h_n)$$

$$= \left. \frac{d}{dt} \right|_{t=0} f(x_1 + th_1, \dots, x_n + th_n)$$

